TRIPATHI STUDY ZONE
JEE & NEET
9112409964
9075829726
CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 8

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three sections in the question paper consisting of Physics (Q.no.1 to 25), Chemistry (Q.no.26 to 50) and Mathematics (Q. no.51 to 75). Each section is divided into two parts, Part I consists of 20 multiple choice questions & Part II consists of 5 Numerical value type Questions.
5. There will be only one correct choice in the given four choices in Part I. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice for Part I Questions and zero mark will be awarded for not attempted question. For Part II Questions 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

PART-I (Multiple Choice Questions)

1. For a certain process the molar heat capacity of an ideal gas is found to be $\left(C_v + \frac{R}{2}\right)$. For the given process it can be concluded that

(a) $PV = \text{constant}$ (b) $\frac{P}{V} = \text{constant}$

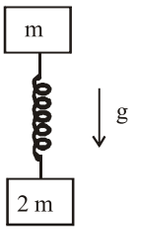
(c) $\frac{V^2}{P} = \text{constant}$ (d) none of these

2. The total pressure P in an ideal gas consisting of diatomic molecules with an internal energy density u is given by

(a) $P = \frac{1}{3}u$ (b) $P = \frac{2}{3}u$

(c) $P = \frac{2}{5}u$ (d) none of these

3. Two bodies of mass m and 2m connected by an unextended spring of spring constant 'k' are allowed to fall simultaneously in a uniform gravitational field g. The extension 'x' in the spring when the bodies A and B are falling



(a) varies with time as $x = A \sin(\omega t + \phi)$

(b) is constant and has value $\sqrt{\frac{2mg}{k}}$

(c) is zero

(d) increases linearly with time

4. An object is moving towards a concave mirror of focal length 24 cm. When it is at a distance of 60 cm from the mirror its speed is 9 cm/s. The speed of its image at the instant is

Space for Rough Work

(a) 4 cm/s towards the mirror
(b) 9 cm/s towards the mirror
(c) 4 cm/s away from the mirror
(d) 9 cm/s away from the mirror

5. The space between the parallel plates of a capacitor is tightly filled with three dielectric slabs A,B, C of thickness 5mm, 3mm and 2mm with dielectric constants 2, 3 and 5 respectively. If a potential difference of 351 volts is applied to plates then electric field intensity in slab A is

(a) 4.5×10^4 V/m (b) 3×10^4 V/m
(c) 2.5×10^4 V/m (d) none of these

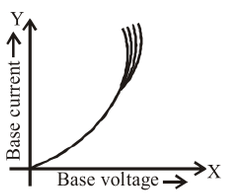
6. Given that radius of earth is 6.4×10^6 m and a transmitting antenna at the top of a tower has a height 40 m and the height of the receiving antenna is 50 m. What is the maximum distance between them for satisfactory communication in los mode?

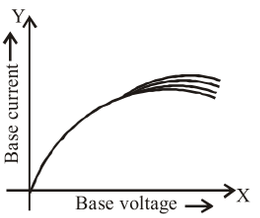
(a) 45 km (b) 48 km (c) 50 km (d) 57 km

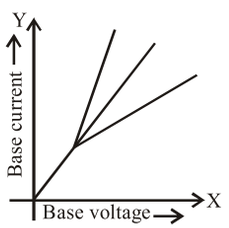
7. While measuring the length of the rod by vernier callipers the reading on main scale is 6.4 cm and the eight division on vernier is in line with marking on main scale division. If the least count of callipers is 0.01 and zero error –0.04 cm, the length of the rod is

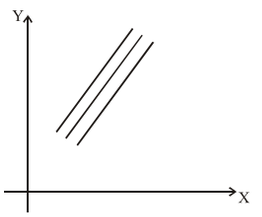
(a) 5.25 cm (b) 6.52 cm (c) 5.52 cm (d) 6.44 cm

8. Which of the following graph represents the input characteristics of a common emmiter transistor correctly ?

(a)


(b)


(c)


(d)


9. Choose the wrong statement from the followings

(a) a capacitor is a two terminal device which conducts with both D.C. and A.C. voltages.
(b) a resistor is a two terminal device which conducts with both D.C. and A.C. voltages.
(c) a diode is a two terminal device which conducts only when it is forward biased.
(d) a transistor is a three terminal device.

10. The figure of merit of a galvanometer is defined as

(a) the voltage required to produce unit deflection in the galvanometer
(b) the current required to produce unit deflection in the galvanometer
(c) the power required to produce unit deflection in the galvanometer
(d) none of these

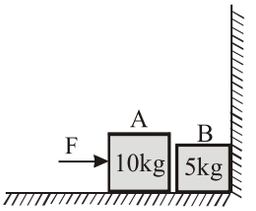
11. The activity of a radioactive element decreases to one-third of the original activity I_0 in a period of nine years. After a further lapse of nine years its activity will be

(a) I_0 (b) $\frac{2}{3}I_0$ (c) $\frac{I_0}{9}$ (d) $\frac{I_0}{6}$

12. In the X-rays tube before striking the target we accelerate the electrons through a potential difference of U volt. For which of the following value of U, we will have X-rays of largest wavelength?

(a) 10 kV (b) 20 kV (c) 30 kV (d) 40 kV

13. Two bodies A and B of masses 10 kg and 5 kg are placed very slightly separated as shown in figure.



The coefficients of friction between the floor and the blocks are as $\mu_s = \mu_k = 0.4$ g between block A and ground breaks, block A will start pressing block B and when welding of B also breaks, block B will start pressing the vertical wall. If F = 20 N, with how much force does block A presses the block B –

(a) 10 N (b) 20 N (c) 30 N (d) Zero

14. A moving particle of mass *m*, makes a head on elastic collision with another particle of mass 2*m*, which is initially at rest. The percentage loss in energy of the colliding particle on collision, is close to

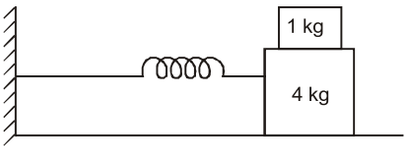
(a) 33% (b) 67% (c) 90% (d) 10%

Space for Rough Work

15. The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10 cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3. The time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbon dioxide will be close to ($In\,5 = 1.601$, $In\,2 = 0.693$).

(a) 231 s (b) 208 s (c) 161 s (d) 142 s

16. The friction coefficient between two blocks of masses 1 kg and 4 kg as shown in figure is μ. The lower plane is smooth. If the system is slightly displaced and released, it will execute SHM. If the upper block does not slip relative to lower block, then maximum amplitude of oscillation should be?



(a) $\frac{5\mu g}{k}$ (b) $\frac{\mu g}{k}$ (c) $\frac{3\mu g}{k}$ (d) $\frac{2\mu g}{k}$

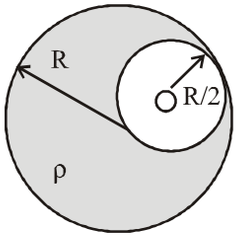
17. Two wave forms, $\psi_1(x, t) = A\sin(kx - \omega t)$ and

$\psi_2(x, t) = A\sin\left(kx + \omega t + \frac{\pi}{3}\right)$ travelling along the x-axis, are superposed. The position of the nodes is given by $\left(k = \frac{2\pi}{\lambda},\ \omega = 2\pi f\right)$

(a) $x_n = \left(n + \frac{1}{6}\right)\frac{\lambda}{2}$ (b) $x_n = \left(n - \frac{1}{6}\right)\frac{\lambda}{2}$

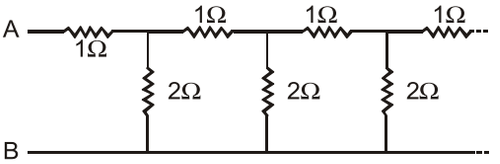
(c) $x_n = \frac{n\lambda}{2}$ (d) none of these

18. There is a spherical cavity of radius R/2 in uniformly charged spherical region having charge density ρ and radius R. A small charged particle having q_0 charge is released at point O. It will collide with the wall of cavity with kinetic energy



(a) $\frac{\rho R^2 q_0}{3\epsilon_0}$ (b) $\frac{\rho R^2 q_0}{6\pi\epsilon_0}$ (c) $\frac{\rho R^2 q_0}{4\pi\epsilon_0}$ (d) $\frac{\rho R^2 q_0}{12\epsilon_0}$

19. The effective resistance between points A and B of infinite network is



(a) 1Ω (b) 2Ω (c) 3Ω (d) 4Ω

20. It takes 15 minutes to raise a certain amount of water from 0°C to boiling point using a heater. Then it takes 1 hour and 20 minutes more to convert all the water into vapour. Then the latent heat of vaporisation of water is

(a) 545 cals (b) 540 cals (c) 533 cals (d) 513 cals

PART-II (Numerical Answer Questions)

21. In an organ pipe, three successive resonance frequencies are observed at 425, 595 and 765Hz. The length (in m) of organ pipe is (Velocity of sound in air = 340 m/s).

22. An aluminium foil of relative emittance of 0.1 is placed in between two concentric spheres at temperature 300 K and 200 K respectively. What is the temperature (in kelvin) of this foil when steady state is reached? Assume the spheres to be black bodies. $\sigma = 5.672 \times 10^{-8}$

23. A change of 0.5 mA in the plate current of a triode occurs when the plate potential is changed by 12 V. If the amplification factor is 16, find the change in grid voltage (in V) required to produce a change of 4 mA in the anode current

24. A person climbs up a stalled escalator in 60 s. If standing on the same but escalator running with constant velocity he takes 40 s. How much time (in second) is taken by the person to walk up the moving escalator?

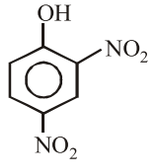
25. In young's double slit experiment, if intensity of maxima on screen is I (assume equal intensity of the source). If the sources become incoherent, average intensity on the screen is I_2, then the ratio $\frac{I_1}{I_2}$ is

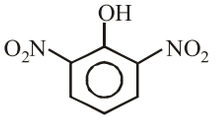
CHEMISTRY

PART-I (Multiple Choice Questions)

26. When temperature is increased, surface tension of water
(a) Increases (b) Remains constant
(c) Decreases (d) Shows irregular behaviour

27. Which of the following does not give effervescence of CO_2 with aq. $NaHCO_3$ solution?

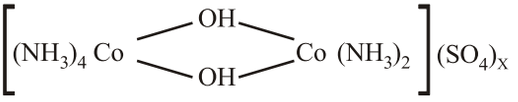
(a) H_2CO_3 (b) 

(c)  (d) None of these

28. Which of the following has highest boiling point?
(a) H_2Se (b) H_2Te (c) H_2S (d) H_2O

29. Which of the following is non-aromatic?
(a) Benzene
(b) Cycloheptatrienyl cation
(c) Cyclopentadienyl anion
(d) Cyclopentadienyl free radical

30. What is the value of x in the following complex?



Consider the oxidation state of Cobalt in the complex same as that of iron in $K_3[Fe(CN)_6]$?
(a) 1 (b) 2 (c) 3 (d) 0

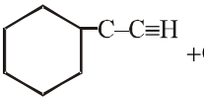
31. Cumene hydroperoxide on hydrolysis yields
(a) Phenol (b) Acetone
(c) Both (d) None of these

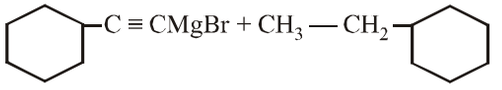
32. Reduction of esters with sodium and alcohol is referred to as
(a) Bouvealt-Blanc reaction
(b) Merdius reaction
(c) Clemensen's reduction
(d) MPV reduction

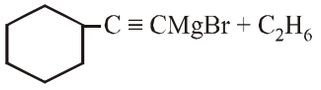
33. *p*-Chloroaniline and anilinium hydrochloride can be distinguished by
(a) Sandmeyer reaction (b) $NaHCO_3$
(c) $AgNO_3$ (d) Carbylamine test

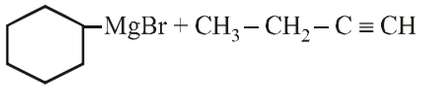
34. In the reaction $I_2 + I^- \longrightarrow I_3^-$ the Lewis acid is
(a) I_2 (b) I^-
(c) I_3^- (d) None of these

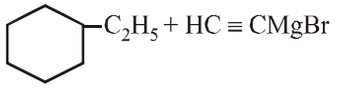
35. Which is the only compound which is deactivating yet *ortho-para* directing when attached to benzene ring?
(a) NO_2 (b) CH_3
(c) Cl_2 (d) None of these

36.  $+CH_3 - CH_2MgBr$ gives

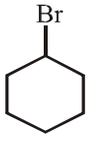
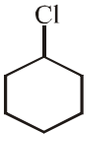
(a) 

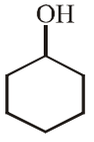
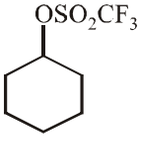
(b) 

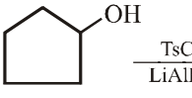
(c) 

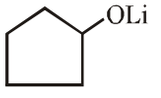
(d) 

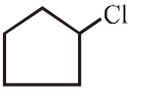
37. Which of the following will be most reactive towards nucleophillic substitution?

(a)  (b) 

(c)  (d) 

38.  A, Then A is

(a)  (b) 

(c)  (d) 

39. The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The potential of the cell would be
(a) 0.177 V (b) −0.177 V (c) 0.087 V (d) 0.059 V

Space for Rough Work

40. The equlibrium const. for reaction is 10. ΔG° will be (R = 8 J/K, T = 300K)

(a) –5527 kJ/mol (b) –5.527 kJ/mol

(c) –55.27 kJ/mol (d) 5.527 kJ/mol

41. In what manner will increase of pressure effect the following equilibrium?

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

(a) Shift in forward direction

(b) Shift in reverse direction

(c) Increase in yield of hydrogen

(d) No effect

42. AgBr dissolves in $Na_2S_2O_3$ solution and forms (A). On boiling an aquous solution of (A), a ppt. (B) is obtained. The colour of B is

(a) Red (b) White (c) Black (d) Colourless

43. A laboratory reagent imparts green colour to the flame. On heating with solid $K_2Cr_2O_7$ and conc. H_2SO_4 it evolves a red gas. Identify the reagent

(a) $CaCl_2$ (b) $BaCl_2$

(c) $CuCl_2$ (d) None of these

44. $KMnO_4$ reacts with oxalic acid as :

$$MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$$

Hence, 50 mL of 0.04 M $KMnO_4$ is acidic medium is chemically equivalent to

(a) 100 mL of 0.1 M $H_2C_2O_4$

(b) 50 mL of 0.2 M $H_2C_2O_4$

(c) 50 mL of 0.1 M $H_2C_2O_4$

(d) 25 mL of 0.1 M $H_2C_2O_4$

45. An alloy of copper, silver and gold is found to have copper constituting the ccp lattice. If silver atoms occupy the edge centres and gold is present at body centre the alloy has a formula

(a) Cu_4Ag_2Au (b) Cu_4Ag_4Au

(c) Cu_4Ag_3Au (d) CuAgAu

PART-II (Numerical Answer Questions)

46. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

At 227°C, heat enthalpy for the above reaction is –22.40 kcal. If gas constant is 1.98×10^{-3} k cal K^{-1} mol^{-1}, find enthalpy at constant volume in k cal mol^{-1} K^{-1}

47. Cyclohexanol is dehydrated to cyclohexene on heating with conc. H_2SO_4. If the yield of this reaction is 75%, how much cyclohexene in grams will be obtained from 100 g of cyclohexanol?

48. If the energy of an electron moving in orbit no. 1 is given by 'E', and energy of an electron in orbit no. 2 of Li^{2+} (z = 3) is E_n. Then calculate the ratio E_n : E.

49. The activation energy for a reaction is 9.0 kcal/mol. Calculate the increase (in %) in the rate constant when its temperature is increased from 298K to 308K.

50. H_2S is a toxic gas used in qualitative analysis. If solubility of H_2S in water at STP is 0.195 m, what is the value of K_H in bar?

MATHEMATICS

PART-I (Multiple Choice Questions)

51. The point on the straight line x + y = 2, which is nearest to the circle $x^2 + y^2 - 10x + 2y + 22 = 0$

(a) (1, 1) (b) (4, –2)

(c) (–2, 9) (d) (3, –1)

52. The function f(x) = 1 + |sin x| is

(a) continuous nowhere

(b) continuous everywhere

(c) continuous only at x = 0

(d) None of these

53. If the length of a focal chord of parabola $y^2 = 4x$ is $\frac{25}{4}$ and has positive slope, then the slope will be

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{4}{3}$ (d) 1

54. Coefficient of x^{25} in the expansion of expression

$$\sum_{r=0}^{50} {}^{50}C_r (2x-3)^r (2-x)^{n-r}$$ is

(a) ${}^{50}C_{25}$ (b) $-{}^{50}C_{24}$ (c) $-{}^{50}C_{25}$ (d) $-{}^{50}C_{30}$

55. The number of ways of selecting 4 cards out of 52 cards of an ordinary pack, so that exactly 3 of them are of same denomination is

(a) 52×48 (b) ${}^{13}C_3 \times {}^{4}C_3 \times 48$

(c) ${}^{52}C_3 \times 48$ (d) ${}^{52}C_3 \times {}^{48}C_3$

Space for Rough Work

56. Let $|\vec{a}| = 2$, $|\vec{b}| = 5$. The values of k for which the vectors $\vec{a} + k\vec{b}$ and $\vec{a} - k\vec{b}$ are perpendicular is

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) $\frac{2}{25}$

57. Domain of the function

$f(x) = \sin^{-1}[1 + \cos x] + \sqrt{16 - x^2}$ ($[\bullet]$ denotes greatest integer function) is

(a) $[-4, 4]$ (b) $(-4, 4)$ (c) $[0, 2\pi]$ (d) $[-4, 4] - \{0\}$

58. Solution of differential equation $\frac{dy}{dx} = \sin x + 2x$, is

(a) $y = x^2 - \cos x + c$ (b) $y = \cos x + x^2 + c$
(c) $y = \cos x + 2$ (d) $y = \cos x + 2 + c$

59. If $\Delta(x) = \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} = A + Bx + Cx^2 + \ldots$

then B =
(a) 0 (b) 1
(c) 2 (d) None of these

60. If in a ΔABC, $\angle C = 90°$ then value of $\tan(A - B)$ is

$\frac{a^2 + b^2}{2ab}$ $\frac{|a^2 - b^2|}{2ab}$

(a) $\frac{|a^2 + b^2|}{2ab}$ (b) $\frac{|a^2 - b^2|}{2ab}$ (c) $\frac{c^4}{2ab}$ (d) $\frac{ca}{2b}$

61. If $z = x + iy$ and $w = \frac{(1 - iz)}{(z - i)}$, then $|w| = 1$, show that in complex plane
(a) z is situated on imaginary axis
(b) z is situated on real axis
(c) z is situated on unit circle
(d) None of these

62. If z, ωz and $\overline{\omega} z$ are the vertices of a triangle, then the area of the triangle will be (where ω is cube root of unity) :

(a) $\frac{3|z|^2}{2}$ (b) $\frac{3\sqrt{3}|z|^2}{2}$

(c) $\frac{\sqrt{3}|z|^2}{2}$ (d) None of these

63. The curve $y = x + \sin x$ has
(a) No maximum point (b) Some maximum points
(c) Some minimum points (d) None of these

64. If $\int \frac{1}{1 + \sin x} dx = \tan\left(\frac{x}{2} + a\right) + b$ then

(a) $a = -\frac{\pi}{4}$, $b \in \mathbf{R}$ (b) $a = \frac{\pi}{4}$, $b \in \mathbf{R}$

(c) $a = \frac{5\pi}{4}$, $b \in \mathbf{R}$ (d) None of these

65. If $f(x) = \begin{cases} \frac{1 - [x]}{1 + x}, & x \neq -1 \\ 1, & x = -1 \end{cases}$, then the value of $f(|2k|)$ will be

(where [] shows the greatest integer function]
(a) Continuous at $x = -1$
(b) Continuous at $x = 0$
(c) Discontinuous at $x = \frac{1}{2}$
(d) All of these

66. A variable circle passes through the fixed point $A(p, q)$ and touches x-axis. The locus of the other end of the diameter through A is
(a) $(y - q)^2 = 4px$ (b) $(x - q)^2 = 4py$
(c) $(y - p)^2 = 4qx$ (d) $(x - p)^2 = 4qy$

67. The orthocentre of the triangle formed by the points A (iz+z), B(z – iz), and C(0) is
(a) 0 + 0i (b) 1 + i (c) 2 + i (d) 3 + 2i

68. The complete solution set for $\sin^{-1}(x) = 3\sin^{-1}(a)$ is

(a) $0 \le a \le \frac{1}{2}$ (b) $-\frac{1}{2} \le a \le 0$

(c) $-\frac{1}{2} \le a \le \frac{1}{2}$ (d) None of these

69. If $\tan^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right) = e^a$ where a is constant, then $\frac{dy}{dx}$ is

(a) $\frac{x}{y}$ (b) $\frac{y}{x}$ (c) $\frac{x^2}{y^2}$ (d) $\frac{y^2}{x^2}$

70. Equation of any plane containing the line

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

is $A(x-x_1)+B(y-y_1)+C(z-z_1)=0$, where

(a) $\frac{A}{a} = \frac{B}{b} = \frac{C}{c}$ (b) $Ax_1 + By_1 + Cz_1 = 0$

(c) $ax_1 + by_1 + cz_1 = 0$ (d) $Aa + Bb + Cc = 0$

PART-II (Numerical Answer Questions)

71. If $x\cos\theta = y\cos\left(\theta+\frac{2\pi}{3}\right) \ge \cos\left(\theta+\frac{4\pi}{3}\right)$, then value of $xy + yz + zx$ is

72. The roots of the equation $x^5 - 40x^4 + px^3 + qx^2 + rx + s = 0$ are in geometric progression and the sum of their reciprocal is 10. Then $|S|$ is equal to

73. If $f(A) = 3$, $f'(A) = -2$, $g(A) = -1$, $g'(A) = 4$, then

$$\lim_{x\to a} \frac{g(x)f(a) - g(a)f(x)}{x-a}$$

74. From a point (h, 0) common tangents are drawn to circles $x^2 + y^2 = 1$ and $(x-2)^2 + y^2 = 4$, value of $|h|$ is

75. If a coin is tossed n times, the probability that head appears odd no. of times is

RESPONSE SHEET

PHYSICS

1.	ⓐⓑⓒⓓ	2.	ⓐⓑⓒⓓ	3.	ⓐⓑⓒⓓ	4.	ⓐⓑⓒⓓ	5.	ⓐⓑⓒⓓ
6.	ⓐⓑⓒⓓ	7.	ⓐⓑⓒⓓ	8.	ⓐⓑⓒⓓ	9.	ⓐⓑⓒⓓ	10.	ⓐⓑⓒⓓ
11.	ⓐⓑⓒⓓ	12.	ⓐⓑⓒⓓ	13.	ⓐⓑⓒⓓ	14.	ⓐⓑⓒⓓ	15.	ⓐⓑⓒⓓ
16.	ⓐⓑⓒⓓ	17.	ⓐⓑⓒⓓ	18.	ⓐⓑⓒⓓ	19.	ⓐⓑⓒⓓ	20.	ⓐⓑⓒⓓ
21.		22.		23.		24.		25.	

CHEMISTRY

26.	ⓐⓑⓒⓓ	27.	ⓐⓑⓒⓓ	28.	ⓐⓑⓒⓓ	29.	ⓐⓑⓒⓓ	30.	ⓐⓑⓒⓓ
31.	ⓐⓑⓒⓓ	32.	ⓐⓑⓒⓓ	33.	ⓐⓑⓒⓓ	34.	ⓐⓑⓒⓓ	35.	ⓐⓑⓒⓓ
36.	ⓐⓑⓒⓓ	37.	ⓐⓑⓒⓓ	38.	ⓐⓑⓒⓓ	39.	ⓐⓑⓒⓓ	40.	ⓐⓑⓒⓓ
41.	ⓐⓑⓒⓓ	42.	ⓐⓑⓒⓓ	43.	ⓐⓑⓒⓓ	44.	ⓐⓑⓒⓓ	45.	ⓐⓑⓒⓓ
46.		47.		48.		49.		50.	

MATHEMATICS

51.	ⓐⓑⓒⓓ	52.	ⓐⓑⓒⓓ	53.	ⓐⓑⓒⓓ	54.	ⓐⓑⓒⓓ	55.	ⓐⓑⓒⓓ
56.	ⓐⓑⓒⓓ	57.	ⓐⓑⓒⓓ	58.	ⓐⓑⓒⓓ	59.	ⓐⓑⓒⓓ	60.	ⓐⓑⓒⓓ
61.	ⓐⓑⓒⓓ	62.	ⓐⓑⓒⓓ	63.	ⓐⓑⓒⓓ	64.	ⓐⓑⓒⓓ	65.	ⓐⓑⓒⓓ
66.	ⓐⓑⓒⓓ	67.	ⓐⓑⓒⓓ	68.	ⓐⓑⓒⓓ	69.	ⓐⓑⓒⓓ	70.	ⓐⓑⓒⓓ
71.		72.		73.		74.		75.	